Business Requirement Document

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Domain	E-commerce/ Retail	Tool Stack	SQL, Power BI, Dax

Executive Summary

This Business Requirements Document outlines the analytical objectives, data sources, methodology, and deliverables for the Olist E-Commerce Business Performance Analysis — a portfolio-grade, end-to-end analytics project built on 100,000+ real Brazilian e-commerce transactions. The project simulates the work of a data analyst embedded in a marketplace business, tasked with surfacing actionable intelligence across revenue, customers, sellers, categories, and logistics.

The output is a 7-page interactive Power BI dashboard backed by 9 advanced SQL queries, a star-schema data model, 12+ DAX measures, and a Recommendations page containing a prioritised Execution Roadmap for stakeholder decision-making.

Business Problem Statement

Olist is a Brazilian B2B2C marketplace connecting thousands of independent sellers to major e-commerce channels. Despite processing 100K+ annual orders, the business lacked a structured analytical view to answer three critical questions:

- **Customer Retention Risk:** Who are the most valuable customers, and is the business retaining them? Initial data revealed that 97% of customers ordered only once — making repeat purchase conversion the single biggest revenue opportunity.
- **Revenue & Category Concentration:** Which product categories and sellers are truly driving profitability? Without this, marketing and inventory decisions lack direction.
- **Logistics Inefficiency:** Where is the delivery experience breaking down, and how does it vary by region? A 3x delivery time gap between São Paulo (8.7 days) and northern states (29+ days) was identified as a direct threat to customer satisfaction.

Business Objective

- Build a full analytics pipeline — from raw relational data in MySQL to an interactive Power BI dashboard.

- Answer specific business questions through advanced SQL queries.
- Design a star-schema data model in Power BI to enable dynamic, cross-filtered analysis.
- Surface actionable insights on revenue trends, customer segments, seller performance, category profitability, and logistics efficiency.
- Deliver a Recommendations page with a prioritised Execution Roadmap for stakeholder use.

Scope

In Scope

- All delivered orders (96,478 records) from the Olist dataset spanning September 2016 to August 2018.
- 7 of 9 relational tables loaded into MySQL (order_reviews excluded due to character encoding issues; geolocation data is redundant).
- SQL business queries covering revenue, segmentation, seller analysis, payment behaviour, and logistics.
- 7-page Power BI dashboard with dynamic filtering by State, Year, and Category.
- DAX measures, star-schema data model, and a Recommendations page.

Out of Scope

- Predictive modelling or machine learning.
- Real-time data integration or live database connections.
- order_reviews table (excluded due to special character import errors in MySQL).

Analytical Workflow

- **Step 1 — Business Questions:** Define the questions stakeholders need answered and map them to measurable KPIs.
- **Step 2 — Data Preparation:** Establish the relevant relationships and prepare the data model for analysis.
- **Step 3 — Targeted SQL Analysis:** Query business-specific questions to validate trends, compare segments and investigate anomalies.
- **Step 4 — Power BI Modelling:** Build the reporting model, calculated measures and interactive report experience.
- **Step 5 — Insight Validation:** Cross-check key dashboard outputs against underlying data and analytical queries.
- **Step 6 — Business Interpretation:** Convert findings into implications, opportunities and risks.
- **Step 7 — Recommendations:** Prioritise actions using the Recommendations page and an execution roadmap.

DATA SOURCES & SCHEMA

Source: Brazilian E-Commerce Public Dataset by Olist — Kaggle.

Table	Rows	Key Columns
orders	99,441	order_id, customer_id, order_status, timestamps
order_items	112,650	order_id, product_id, seller_id, price, freight
customers	99,441	customer_id, customer_unique_id, state, city
order_payments	103,886	order_id, payment_type, installments, value
products	32,951	product_id, category_name, dimensions
sellers	3,095	seller_id, seller_state, seller_city
category_translation	69	product_category_name (PT $\rightarrow$ EN)
order_reviews	*Skipped*	review_score, comments

SQL Business Queries

1. MONTHLY REVENUE TREND

   Business Problem: Is the business growing or declining? Where are the seasonal peaks we should plan for?

2. TOP REVENUE CATEGORIES

   Business Problem: Which product categories should we prioritise for marketing and inventory investment?

3. TOP STATES BY ORDERS & REVENUE

   Business Problem: Which regions drive the most value? Where should we expand or reduce operations?

4. BEST & WORST PERFORMING SELLERS

   Business Problem: Which sellers are driving revenue and which are underperforming on the platform?

5. PAYMENT METHOD ANALYSIS

   Business Problem: How are customers paying and does payment behavior vary by order value?

6. CUSTOMER SPEND SEGMENTATION
   Business Problem: Are we over-relying on a small group of high-value customers for most of our revenue?
7. RFM SEGMENTATION
   Business Problem: Who are our most valuable customers and who is at risk of churning?
8. AVERAGE DELIVERY TIME BY STATE
   Business Problem: Which states have logistics problems? Are we meeting our promised delivery dates?
9. SELLER CONCENTRATION RISK
   Business Problem: Is our revenue dangerously dependent on a few top sellers or is it healthily distributed?

Power BI Report

A star-schema was designed in Power BI with order_items as the central fact table, connected to dimensions: orders, customers, products, sellers, category_translation, order_payment, and a custom Calender table named as Dim_Date.

Dashboard Pages

Page	Title	Business Question Answered
1	Executive Overview	What is the overall business performance?
2	Sales & Revenue	Where is revenue coming from and how is it growing?
3	Customer Intelligence	Who are our customers and where are they?
4	Seller & Supply Chain	Which sellers are performing and where is concentration risk?
5	Product & Category Insights	Which categories drive revenue vs volume vs price?
6	Logistics & Delivery	Where are deliveries failing and how bad is the gap?
7	Recommendations	What should the business do next?

Success Criteria

- All seven Power BI pages are complete and mapped to defined business questions.
- Core KPIs are calculated consistently and validated.
- Interactive filters operate correctly across applicable pages.
- Key findings can be traced to underlying data and analytical logic.
- Report design communicates insights clearly to a non-technical stakeholder.
- Recommendations are supported by observed findings rather than unsupported assumptions.
- The GitHub documentation explains the business problem, data, model, dashboard structure and findings.